

MINI PROJECT REPORT

on

**Light Sensing Security System(I.R PHOTO
DIODE BASED CIRCUIT)**

Submitted by

B.V.BALANILAVAN-RA231105301023

P .SUKESH-RA2311053010127

K.HEMANTH-RA2311053010128

KESHAV B.S -RA2311053010144

Semester – II

Academic Year: 2023-24 Even

Under the guidance of

Dr. A.Manikandan

Assistant Professor,

Department of ECE

In partial fulfilment for the Course

of

21ECC101J-Electronic System and PCB Design



**DEPARTMENT OF ELECTRONICS AND
COMMUNICATION ENGINEERING**

College of Engineering and Technology,

SRM Institute of Science and Technology

SRM Nagar, Kattankulathur – 603203, Kancheepuram District, Tamil Nadu.

April 2024

Introduction:

An infrared (IR) photodiode-based light-sensing security system serves as an effective and straightforward application of electronic sensors in safety and security contexts. This system uses a simple electronic circuit to detect changes in IR light levels caused by an obstruction (such as a person passing through a doorway) and triggers an alert via a buzzer and an LED indicator. The design is compact, cost-effective, and can be assembled using commonly available electronic components.

Objective:

The objective is to build a light-sensing security system using an IR photodiode to detect changes in light levels and trigger alerts, serving both educational and practical purposes.

Components:

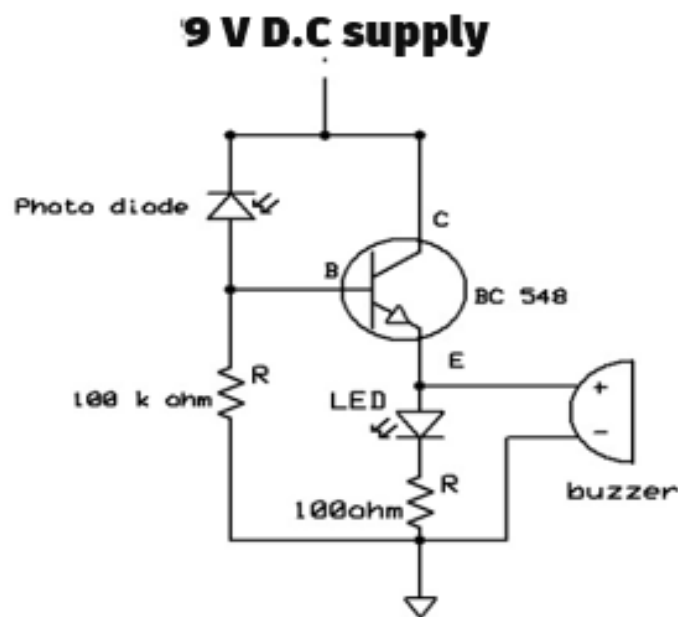
- 5V DC battery
- Battery connector
- Photo diode
- LED
- Buzzer
- 100k ohm resistor
- 100 ohm resistor
- Transistor BC 548
- PCB breadboard-MB102

Background and Theory:

Traditional fire detection methods have limitations in providing early and accurate detection. Infrared (IR) photodiodes offer a promising solution due to their sensitivity to infrared radiation emitted by flames. These photodiodes operate based on the principle of photoconductivity, where exposure to infrared radiation increases conductivity, resulting in a change in current flow.

Our project utilizes IR photodiodes in a transistor-based circuit to detect infrared radiation from fires. This circuit amplifies the photodiode signal and triggers visual and auditory alarms, ensuring timely notification of light incidents.

Circuit Design:



Circuit Operation:

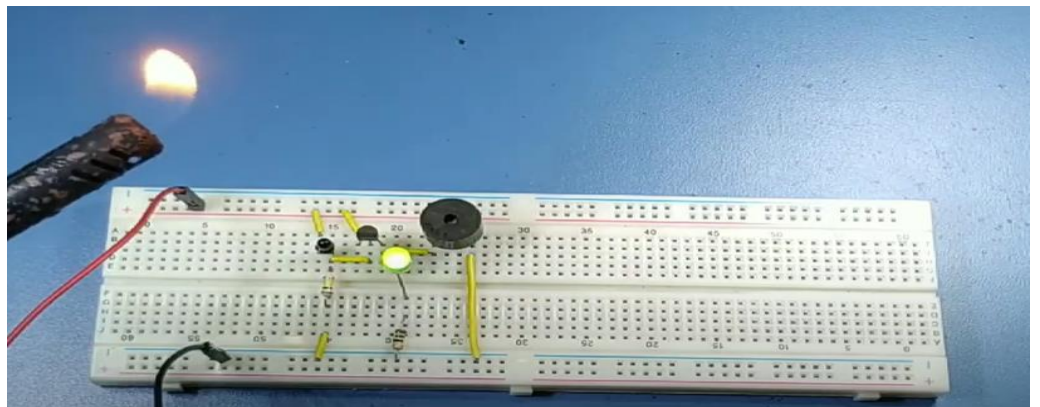
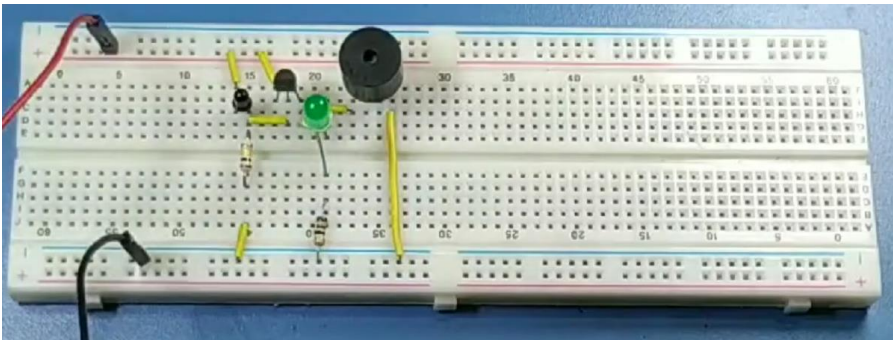
- 1. Detection Phase:** The IR photodiode is sensitive to the IR radiation from flames. Upon exposure to such radiation, its conductivity increases, allowing more current to flow through it.
- 2. Signal Amplification:** The increased current from the photodiode flows into the base of the transistor (BC 548). The transistor acts as a switch; when it receives sufficient base current, it conducts between its collector and emitter, activating the connected load (the LED and buzzer).
- 3. Alerting:** When the transistor conducts, the circuit closes, powering the LED and buzzer. The LED lights up, and the buzzer sounds an alarm, indicating that a fire has been detected.

Methodology:

Our methodology involves systematic steps from circuit design to testing and refinement:

- 1) **Circuit Design:** Components are carefully selected and a schematic is drawn.
- 2) **Circuit Assembly:** Components are mounted on the PCB with precise soldering.
- 3) **Programming and Configuration:** Circuit parameters are adjusted for optimal performance.
- 4) **Testing:** Functional, performance, and reliability tests are conducted.
- 5) **Evaluation and Refinement:** Data analysis guides adjustments to improve circuit performance.
- 6) **Documentation:** Detailed documentation is prepared, including a user manual for operation.

Fabricated pcb assembly image and its output measurement



Conclusion:

In conclusion, the infrared (IR) photodiode-based light-sensing security system offers a simple yet effective solution for detecting disturbances in a monitored area. By leveraging basic electronic components and principles, the system demonstrates the functionality of light sensors and provides a cost-effective security option for various applications. Its educational value lies in its ability to teach basic electronics concepts, while its practicality makes it suitable for home or small business security needs. With further enhancements and customization, this system can be adapted for broader security applications.